AlloQuant master results - data dictionary

File: master_kinase_analysis_results_v7r2.csv | 1600 rows x 48 columns | generated 2026-07-21

- This dictionary defines the AlloQuant v7r2 master-CSV SCHEMA. The 53 columns are identical across all v7r2 masters (SRC 260606, SRC 260718 rerun, CDK1-CCNB1); only which rows are populated varies by run.
- Anchored to the SRC 260718 rerun (the AF3-complete master). Empirical counts/ranges below are from that file: 8 systems (CSK apo/holo x SRC apo/holo x SRC WT/pY159) x 25 AF3 models x 2 chains = 1600 rows.
- AF3 confidence columns (ipTM/pTM/PAE_*) are populated in this rerun (it was re-predicted to capture summary_confidences.json/confidences.json). They are N/A in the original SRC 260606 master. The CDK1 master is single-chain (kinase only, 600 rows) with AF3 columns N/A for its monomer models.
- holo = ATP(/Mg)-bound; apo = no nucleotide. C_Spine == 'No Ligand' flags an apo chain.
- Distances in Angstrom (A), angles in degrees. Landmark indices are 0-based HMM-aligned positions; numbered landmarks (n99, v104, k105, e107, m118, m120, e121, i150, y156, d220) are labelled in CSK numbering and map to the equivalent target position via the HMM.
- Missing / uninformative values: geometry that fails to compute is written as the sentinel 999.9; everything else uninformative is the string 'N/A' (or 'None' for CoFactor_Name / Partner). Downstream R reads '', 'NA', 'N/A', 'None' all as NA.
- Writers: columns 1-48 by modules/chimerax_hmm_worker_v7r2.py; columns 49-53 (ipTM, pTM, PAE_*) by modules/extract_af3_metrics.py, joined on Directory.

#	Column	Group	Type	Units	Populated	Allowed values / range	Empirical (this CSV)	Description
1	**Simulation_ID**	Identity	string		always		filled 1600/1600	Unique model id, from the model path via get_core_name() (dedups repeated path tokens). Encodes the two chains, AF3 seed and sample, e.g. a-csk-wtcat-apo_b-src-wtcat-holo_seed-..._sample-N_model.
2	**Directory**	Identity	string (path)		always		filled 1600/1600	Relative path to the AF3 prediction folder (system / seed-sample). Also the join key used to merge in the AF3 confidence columns.
3	**File**	Identity	string		always	model.cif	filled 1600/1600	Structure file analysed within Directory (AF3 predicted mmCIF).
4	**Chain**	Identity	categorical		always	A, B, D	filled 1600/1600	mmCIF chain id of the protein chain in this row. A = CSK protomer; B or D = SRC protomer (the letter shifts to D when holo ligand chains are inserted ahead of it). One row per protein chain; 2 rows per model.
5	**Type**	Identity	categorical		always	CSK-WT, SRC-WT	filled 1600/1600	Kinase identity assigned by best sequence-identity match to the FASTA landmark set (get_best_landmark_for_chain). Note the SRC pY159 variant still matches as SRC-WT; the pY159 vs WT distinction lives in Simulation_ID/Directory, not here.
6	**State**	Conformation	categorical		always	see Description	filled 1600/1600	Human-readable state = Spatial + Dihedral with an activity gloss: Active (BLAminus), Active-Like (ABAminus), Inactive (BLBplus / BLBtrans / BLAplus / BLBminus / BBAminus), else a fall-through <spatial> (<dihedral>). Downstream analysis treats 'Active (BLAminus)' as the active target.
7	**Role**	Interface	categorical		always (dimer)	C-lobe Donor, N-lobe Receiver, Unpaired	filled 1600/1600	Role in the asymmetric dimer from analyze_dimer_interface: a chain is C-lobe Donor / N-lobe Receiver when its C-lobe contacts the partner N-lobe within 8 A (Symmetric if mutual, Unpaired if no contact). For CSK-SRC: CSK=N-lobe Receiver, SRC=C-lobe Donor.
8	**Partner**	Interface	categorical		always (dimer)	A, B, D	filled 1598/1600, blank 2	Chain id of this chain's interface partner (or None if Unpaired; a cofactor name for Co-factor Bound systems).
9	**CoFactor_Name**	Cofactor	string		N/A (no cofactor)		filled 0/1600, blank 1600	Name of the nearest non-kinase chain within 15 A, else None. Populated only for cofactor systems (CDK-cyclin, RAF/14-3-3, etc.); empty for the CSK-SRC pair.
10	**CoFactor_aC_Dist**	Cofactor	float	A	N/A (no cofactor)		filled 0/1600, blank 1600	Min distance from the kinase alphaC region (c-4..c+4) to the cofactor chain. Cofactor systems only.
11	**CoFactor_ActLoop_Dist**	Cofactor	float	A	N/A (no cofactor)		filled 0/1600, blank 1600	Min distance from the kinase activation loop (f..APE) to the cofactor chain. Cofactor systems only.
12	**Interface_C_Lobe_Donor_Dist**	Interface	float	A	always (dimer)	2.44 to 13.24	filled 1598/1600, blank 2	Min distance between chain-A C-lobe and chain-B N-lobe atoms (donor face). 999.9 sentinel if lobes cannot be split; N/A if unpaired.
13	**Interface_N_Lobe_Rec_Dist**	Interface	float	A	always (dimer)	2.44 to 13.24	filled 1598/1600, blank 2	Min distance between chain-B C-lobe and chain-A N-lobe atoms (receiver face). Same sentinel/NA conventions.
14	**R_Spine**	Spine/helix	categorical		always	Intact, Broken	filled 1600/1600	Regulatory-spine integrity: Intact if all three R-spine contacts (HRD-His-DFG-Phe, DFG-Phe-rs1, rs1-rs2) are <4.5 A, else Broken (Missing if landmarks absent).
15	**C_Spine**	Spine/helix	categorical		always	Intact, No Ligand	filled 1600/1600	Catalytic-spine vs adenine: Intact if the VAIK region (k..k-3) is <6 A from a bound ligand, Ligand Distant if >=6, No Ligand if apo. Serves as the apo/holo discriminator downstream (No Ligand = apo chain).
16	**C_Helix**	Spine/helix	categorical		always	In, Out	filled 1600/1600	alphaC rotamer: In if the VAIK-Lys CB to alphaC-Glu CB distance is <=10 A (salt-bridge competent), else Out. In = active-like.
17	**Shell_State**	Spine/helix	categorical		always	Packed	filled 1600/1600	Regulatory-shell packing: Packed if Shell_M118_M120_Dist <5 A, else Loose (only Packed observed in this run).
18	**Spatial**	Conformation	categorical		always	DFGin, DFGout, Outlier	filled 1600/1600	Dunbrack spatial DFG group from D1_Dist/D2_Dist: DFGin (D1<=11 & D2>=11), DFGout (D1>11 & D2<=14), DFGinter (both <=11), else Outlier.
19	**Dihedral**	Conformation	categorical		always	BLAminus, BLBplus, ...	filled 1600/1600	KinCore backbone cluster = Ramachandran regions of X(f-2), Asp(f-1), Phe(f) + DFG-Phe chi1 rotamer, e.g. BLAminus/ABAminus/BLBplus/BLBtrans/BLAplus/BLBminus. Region codes A/B/L/E; rotamer plus/trans/minus. BLAminus = canonical active.
20	**ActLoop_NT**	Conformation	categorical		always	NTin, NTout	filled 1600/1600	Activation-loop N-terminal packing: NTin if residues f+3..f+6 come within 5.5 A of the residue before HRD (hrd-1), else NTout.
21	**ActLoop_CT**	Conformation	categorical		always	CTin, CTout	filled 1600/1600	Activation-loop C-terminal packing: CTin if the pre-APE window comes within 5.5 A of hrd+6 (an Arg), else CTout.
22	**Phi_D**	Conformation	float	degrees	always	-100.82 to 70.18	filled 1600/1600	Backbone phi dihedral of the DFG-Asp (f-1).
23	**Psi_D**	Conformation	float	degrees	always	15.46 to 113.62	filled 1600/1600	Backbone psi dihedral of the DFG-Asp (f-1).
24	**Cleft_Gape_Dist**	Active site	float	A	always	11.65 to 19.61	filled 1600/1600	Inter-lobe cleft opening: roof Calpha to floor Calpha, where roof = a P-loop residue (phospho/Tyr/Thr/Ser or P-loop midpoint) and floor = catalytic HRD-Asp (hrd+2). Geometric, so populated for apo and holo.
25	**Mg_Hijack_Dist**	Active site	float	A	holo + Mg only	7.64 to 64.16	filled 1200/1600, blank 400	Min distance from the roof residue phosphate/hydroxyl atoms to any Mg2+ ion. Requires a bound Mg2+; N/A on apo chains.

#	Column	Group	Type	Units	Populated	Allowed values / range	Empirical (this CSV)	Description
26	**Substrate_Clearance_Angle**	Active site	float	degrees	nucleotide-bound only	88.33 to 158.70	filled 800/1600, blank 800	Angle roof-Calpha / ATP-phosphate / floor-Calpha (phosphate atom as vertex). Requires a bound nucleotide; N/A on apo chains.
27	**D1_Dist**	Active site	float	A	always	8.14 to 12.50	filled 1600/1600	Dunbrack D1 coordinate: alphaC-Glu Calpha to DFG-Phe CZ (farthest side-chain atom if CZ absent).
28	**D2_Dist**	Active site	float	A	always	7.80 to 16.74	filled 1600/1600	Dunbrack D2 coordinate: VAIK-Lys Calpha to DFG-Phe CZ.
29	**SB_Dist**	Active site	float	A	always	2.44 to 18.43	filled 1600/1600	beta3 Lys - alphaC Glu salt bridge: min distance from VAIK-Lys NZ to the nearest polar (O*/N*) side-chain atom of the alphaC-Glu. Short (~2.5-3 A) = intact/active.
30	**HRD_ATP_Dist**	Active site	float	A	holo only	2.47 to 8.92	filled 800/1600, blank 800	Min distance from catalytic HRD-Asp (hrd+2) side-chain O*/N* to the ATP gamma/beta/alpha-phosphate group. Requires a bound nucleotide; N/A on apo chains.
31	**DFG_Mg_Dist**	Active site	float	A	holo + Mg only	0.91 to 6.97	filled 796/1600, blank 804	Min distance DFG-Asp (f-1) side-chain O*/N* to the nearest Mg2+ (recorded only if <=15 A). Requires Mg2+; N/A on apo chains.
32	**DFG_ATP_Dist**	Active site	float	A	holo only	1.26 to 4.70	filled 800/1600, blank 800	Min distance DFG-Asp side-chain O*/N* to ATP phosphate atoms. Requires a bound nucleotide; N/A on apo chains.
33	**PLoop_ATP_Dist**	Active site	float	A	holo only	1.44 to 5.22	filled 800/1600, blank 800	Min distance from P-loop backbone/CB atoms to ligand phosphate atoms. Requires a bound nucleotide; N/A on apo chains.
34	**aCb4_aE_Dist**	Allosteric	float	A	always	2.81 to 3.72	filled 1600/1600	Min polar-atom (N*/O*) distance between the alphaC-beta4 loop (c+8..c+14) and the alphaE helix (hrd-25..hrd-10). Allosteric relay bridge.
35	**Spine_Bridge_Dist**	Allosteric	float	A	always	3.57 to 8.36	filled 1600/1600	alphaC-beta4 loop to ligand if a ligand is bound; otherwise (apo) loop to k-3 residue. Populated for both apo and holo (fallback for apo).
36	**V104_RS2_Dist**	Allosteric	float	A	always	6.81 to 7.38	filled 1600/1600	Min side-chain distance V104 to R-spine residue rs2 (beta4).
37	**I150_HRD_Dist**	Allosteric	float	A	always	9.65 to 10.51	filled 1600/1600	Min side-chain distance I150 to the catalytic HRD-His.
38	**Shell_M118_M120_Dist**	Allosteric	float	A	always	3.57 to 4.45	filled 1600/1600	Min side-chain distance between the two shell methionines M118 and M120 (drives Shell_State).
39	**Y156_N99_Dist**	Allosteric	float	A	always	7.87 to 9.86	filled 1600/1600	Residue-min distance Y156 to N99 (alphaE anchor).
40	**K105_E107_Dist**	Allosteric	float	A	always	2.95 to 5.71	filled 1600/1600	Min side-chain distance K105 to E107.
41	**K105_E121_Dist**	Allosteric	float	A	always	2.97 to 4.16	filled 1600/1600	Min side-chain distance between residues at Pfam Pkinase HMM node 62 (PKA canonical K105 position; GLN65 in CSK domain numbering) and HMM node 78 (PKA canonical E121 position; GLU82 in CSK domain numbering). Part of the alphaC-beta4 loop contact network. NOT the beta3-Lys/alphaC-Glu regulatory salt bridge -- for that, see SB_Dist.
42	**K105_N99_Dist**	Allosteric	float	A	always	2.72 to 7.11	filled 1600/1600	Min side-chain distance K105 to N99.
43	**D220_HRD_Dist**	Allosteric	float	A	always	3.35 to 3.84	filled 1600/1600	Min side-chain distance D220 to the catalytic HRD-His.
44	**ipTM**	AF3 confidence	float	0-1	dimers	0.32 to 0.90	filled 1600/1600	AF3 interface pTM from summary_confidences.json (>0.8 high, 0.6-0.8 moderate, <0.6 low). N/A for single-entity/monomer models. Populated in this rerun; N/A in the original 260606 master.
45	**pTM**	AF3 confidence	float	0-1	always	0.57 to 0.91	filled 1600/1600	AF3 predicted TM-score. Populated in this rerun; N/A in the original 260606 master.
46	**PAE_A_to_B**	AF3 confidence	float	A	dimers	8.40 to 27.45	filled 1600/1600	Mean predicted aligned error over the [chain-A rows, chain-B cols] PAE block (<5 rigid, >15 non-interacting). N/A for monomers.
47	**PAE_B_to_A**	AF3 confidence	float	A	dimers	7.99 to 26.56	filled 1600/1600	Mean PAE over the transposed [chain-B rows, chain-A cols] block. N/A for monomers.
48	**PAE_Mean_AB**	AF3 confidence	float	A	dimers	8.43 to 27.00	filled 1600/1600	Symmetric interface PAE = mean of PAE_A_to_B and PAE_B_to_A. N/A for monomers.